
Program 1: Working with Maven: Creating a Maven Project, Understanding the POM File, Dependency Management and Plugins

Step 1: To create a Maven project run the following command in terminal:

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

This command generates a basic Maven project structure with a sample App.java and AppTest.java.

```
sumanth@sumanth-VirtualBox:~$ cd devops
sumanth@sumanth-VirtualBox:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 6.9 MB/s)
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/sumanth/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/sumanth/devops/MyMavenApp
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 5.299 s
[INFO] Finished at: 2026-03-19T13:15:32+05:30
[INFO]
```

This “BUILD SUCCESS” will confirm that the project is created successfully

```
sumanth@sumanth-VirtualBox:~/devops$ cd MyMavenApp
sumanth@sumanth-VirtualBox:~/devops/MyMavenApp$ ls
pom.xml  src
sumanth@sumanth-VirtualBox:~/devops/MyMavenApp$ gedit pom.xml
```

Step 2: Go to the project directory using command **cd MyMavenApp**

Step 3: Open and edit pom.xml file using command **gedit pom.xml**

- Go to <https://mvnrepository.com>
- Search for a dependency “junit”
- Select the latest version and copy the dependency XML snippet.
- Paste it under the <dependencies> section in pom.xml
- Updated pom.xml:

```

pom.xml
~/Devops/MyMavenApp
Save

1 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
2   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
3   <modelVersion>4.0.0</modelVersion>
4   <groupId>com.example</groupId>
5   <artifactId>MyMavenApp</artifactId>
6   <packaging>jar</packaging>
7   <version>1.0-SNAPSHOT</version>
8   <name>MyMavenApp</name>
9   <url>http://maven.apache.org</url>
10  <!-- Properties: Customize Java version or plugin versions -->
11  <properties>
12    <maven.compiler.source>11</maven.compiler.source>
13    <maven.compiler.target>11</maven.compiler.target>
14  </properties>
15  <dependencies>
16    <dependency>
17      <groupId>junit</groupId>
18      <artifactId>junit</artifactId>
19      <version>3.8.1</version>
20      <scope>test</scope>
21    </dependency>
22  </dependencies>
23  <!-- Build: Configuring plugins and build settings -->
24  <build>
25    <plugins>
26      <!-- Example: Maven Compiler Plugin to compile Java code -->
27      <plugin>
28        <groupId>org.apache.maven.plugins</groupId>
29        <artifactId>maven-compiler-plugin</artifactId>
30        <version>3.8.1</version>
31        <configuration>
32          <source>1.7</source>
33          <target>1.7</target>
34        </configuration>
35      </plugin>
36
37      <!-- Example: Maven Surefire Plugin to run tests -->
38      <plugin>
39        <groupId>org.apache.maven.plugins</groupId>
40        <artifactId>maven-surefire-plugin</artifactId>
41        <version>2.22.2</version>
42      </plugin>
43
44      <plugin>
45        <groupId>org.apache.maven.plugins</groupId>
46        <artifactId>maven-jar-plugin</artifactId>
47        <version>3.1.0</version>
48        <configuration>
49          <archive>
50            <manifestEntries>
51              <Main-Class>com.example.MainClass</Main-Class>
52            </manifestEntries>
53          </archive>
54        </configuration>
55      </plugin>
56    </plugins>
57  </build>
58 </project>

```

Step 4: Build the project using command **mvn clean install**

```

sumanth@sumanth-VirtualBox:~/devops/MyMavenApp$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenApp >-----
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenApp ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/sumanth/devops/MyMavenApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/sumanth/devops/MyMavenApp/target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/sumanth/devops/MyMavenApp/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/sumanth/devops/MyMavenApp/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenApp ---
[INFO]

```

```

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.028 s - in com.example.AppTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenApp ---
[INFO] Building jar: /home/sumanth/devops/MyMavenApp/target/MyMavenApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenApp ---
[INFO] Installing /home/sumanth/devops/MyMavenApp/target/MyMavenApp-1.0-SNAPSHOT.jar to /home/sumanth/.m2/repository/com/example/MyMavenApp/1.0-SNAPSHOT/MyMavenApp-1.0-SNAPSHOT.jar
[INFO] Installing /home/sumanth/devops/MyMavenApp/pom.xml to /home/sumanth/.m2/repository/com/example/MyMavenApp/1.0-SNAPSHOT/MyMavenApp-1.0-SNAPSHOT.pom
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.495 s
[INFO] Finished at: 2026-03-19T13:17:37+05:30
[INFO] -----

```

The “Build Success” message ensures that our Maven project is built successfully.

Step 5: To run the Maven application type the command:

java -jar target/MyMavenApp-1.0-SNAPSHOT.jar

- After completing the execution, the output will look like:

```

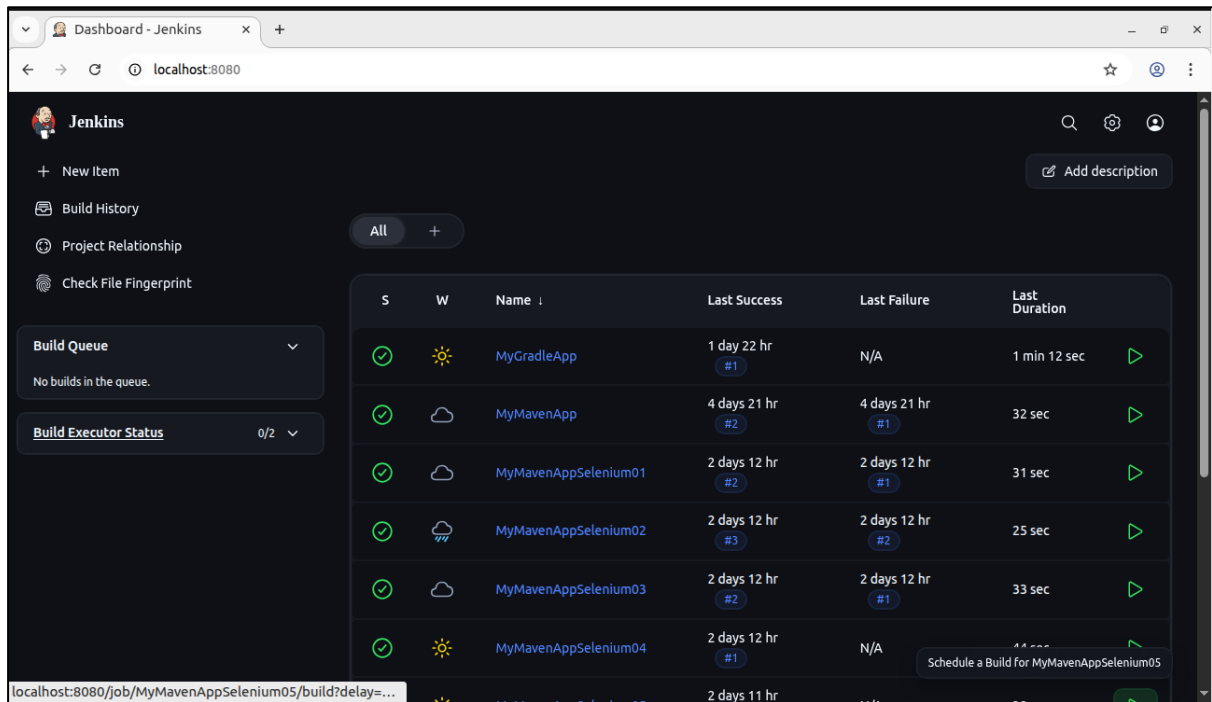
sumanth@sumanth-VirtualBox:~/devops/MyMavenApp$ java -jar target/MyMavenApp-1.0-SNAPSHOT.jar
Hello This is Sumanth from BIT
sumanth@sumanth-VirtualBox:~/devops/MyMavenApp$ █

```

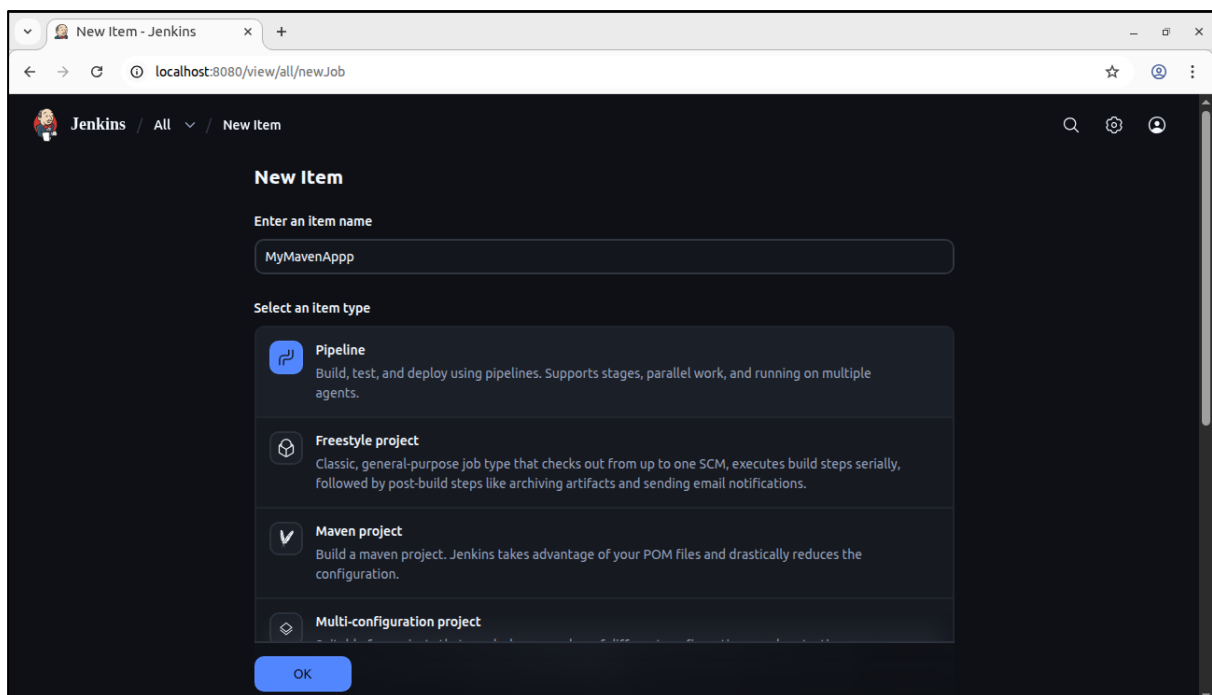
This will show output as the code present in App.java file we can edit the text to get any other output.

Jenkins Pipeline for MyMavenApp

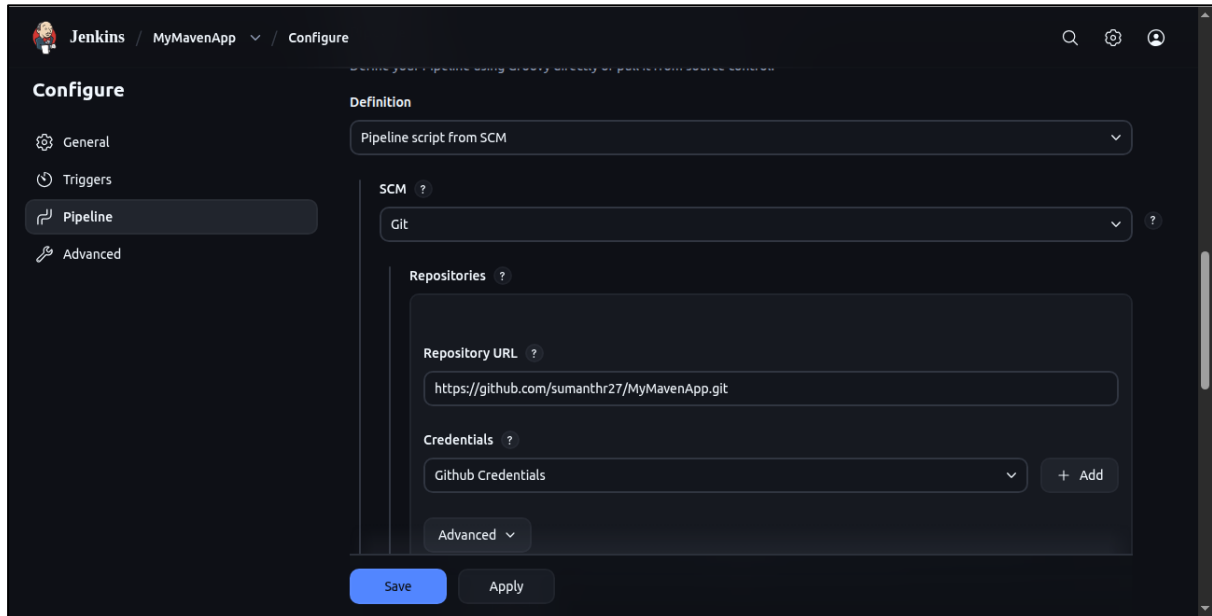
Step 1: Open the Jenkins Dashboard in order to create a new item



Step 2: Click on New Item in order for creating an New Item and thus give the name as “MyMavenApp” and Item type as Pipeline and Create it



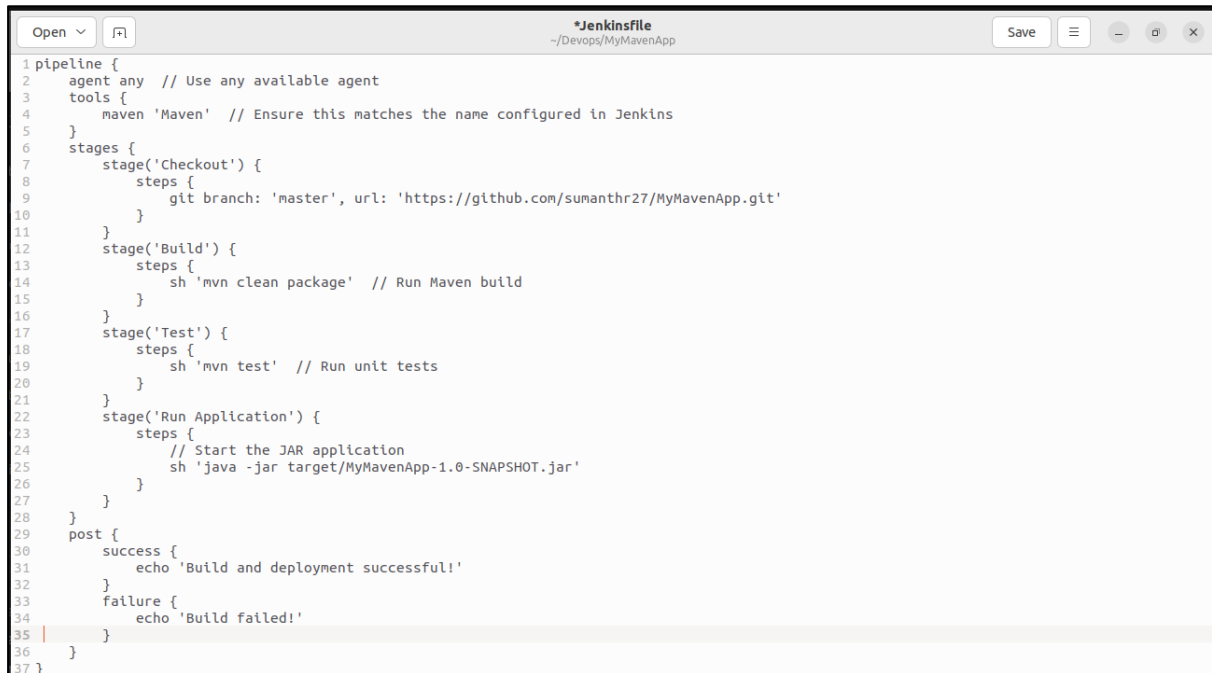
Step 3: Configure the settings by Defining the script as “Pipeline Script from SCM” and SCM as “Git”. Thus, now as we have selected Git, we need to paste the Repository URL of the project given in GitHub and choose Credentials as “GitHub Credentials” which was created earlier.



Step 4: Now in Terminal, go to directory created for MyMavenApp and create a new file called “Jenkinsfile”

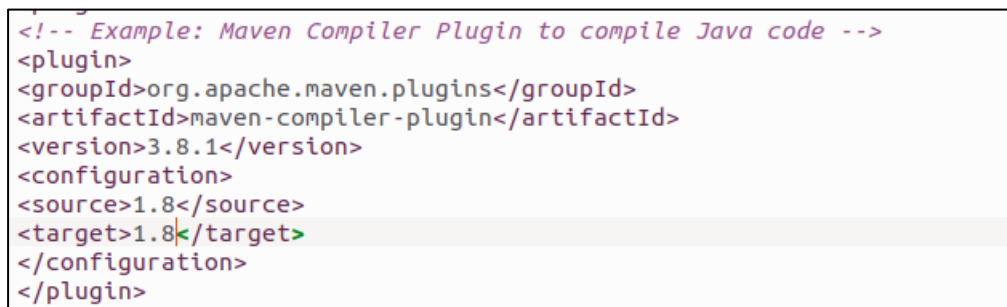
```
sumanth@sumanth-VirtualBox: ~/Devops/MyMavenApp
sumanth@sumanth-VirtualBox:~$ cd devops
bash: cd: devops: No such file or directory
sumanth@sumanth-VirtualBox:~$ cd Devops
sumanth@sumanth-VirtualBox:~/Devops$ cd MyMavenApp
sumanth@sumanth-VirtualBox:~/Devops/MyMavenApp$ gedit Jenkinsfile
```

Step 5: Paste the available code inside the Jenkinsfile



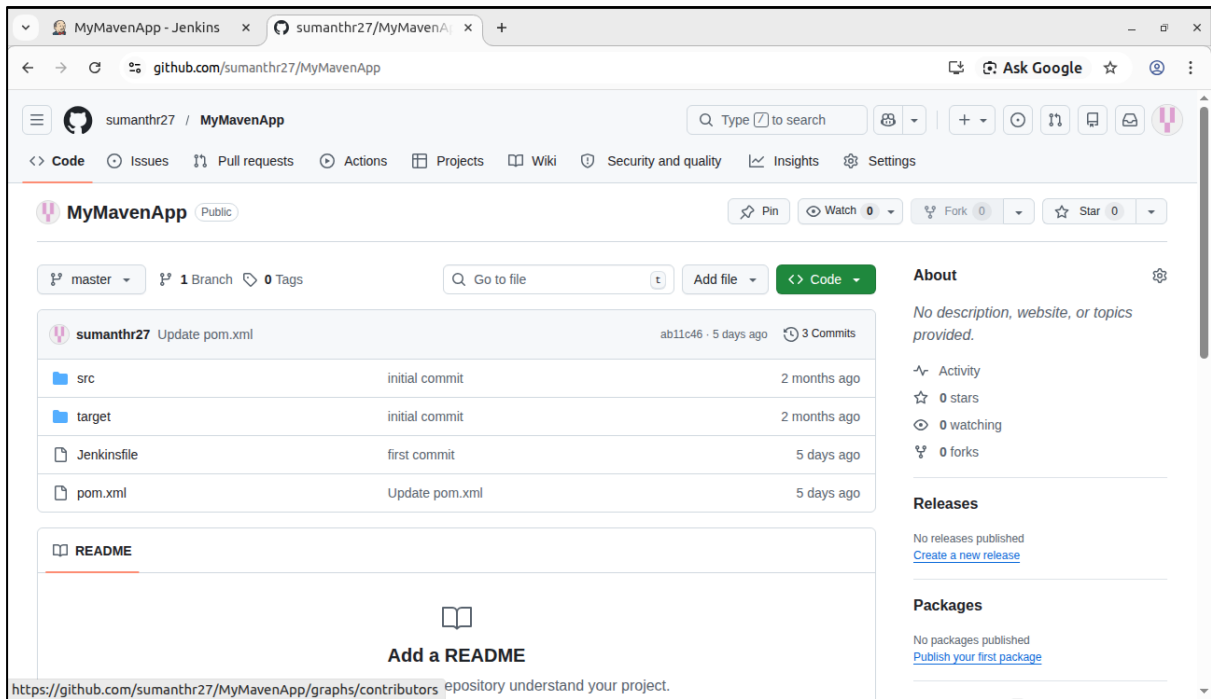
```
1 pipeline {
2   agent any // Use any available agent
3   tools {
4     maven 'Maven' // Ensure this matches the name configured in Jenkins
5   }
6   stages {
7     stage('Checkout') {
8       steps {
9         git branch: 'master', url: 'https://github.com/sumanthr27/MyMavenApp.git'
10      }
11    }
12    stage('Build') {
13      steps {
14        sh 'mvn clean package' // Run Maven build
15      }
16    }
17    stage('Test') {
18      steps {
19        sh 'mvn test' // Run unit tests
20      }
21    }
22    stage('Run Application') {
23      steps {
24        // Start the JAR application
25        sh 'java -jar target/MyMavenApp-1.0-SNAPSHOT.jar'
26      }
27    }
28  }
29  post {
30    success {
31      echo 'Build and deployment successful!'
32    }
33    failure {
34      echo 'Build failed!'
35    }
36  }
37 }
```

Step 6: We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8



```
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
  <groupId>org.apache.maven.plugins</groupId>
  <artifactId>maven-compiler-plugin</artifactId>
  <version>3.8.1</version>
  <configuration>
    <source>1.8</source>
    <target>1.8</target>
  </configuration>
</plugin>
```

Step 7: Now we need to Push changes and commit by pushing the necessary “Jenkinsfile” to the respective Repo



Step 8: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.

